

IndyGo On-Time Performance PowerBI Dashboard

Sudhir Pol
Indiana University Bloomington
sypol@iu.edu

Aditya Kadam
Indiana University Bloomington
askadam@iu.edu

Christopher Okoro
Indiana University Bloomington
cookoro@iu.edu

Mohammed Pathariya
Indiana University Bloomington
mjpathar@iu.edu

Mouryan Jayasankar
Indiana University Bloomington
mjayasan@iu.edu

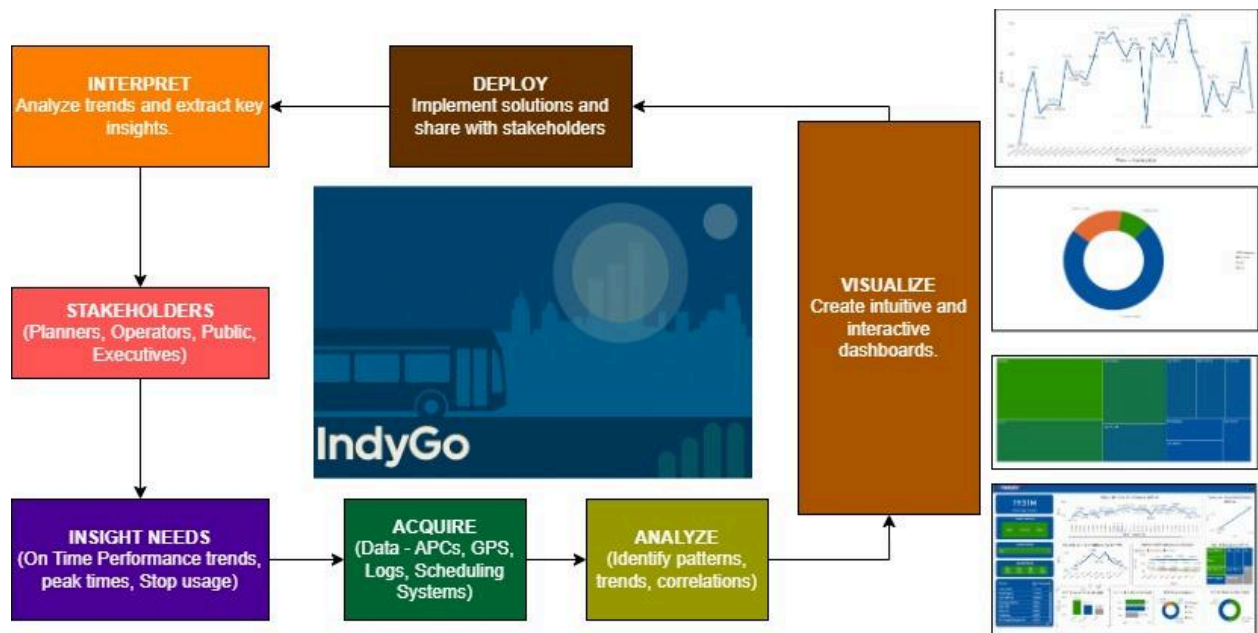


Fig. 1. On Time Performance Data Flow: From Stakeholder Insights to Decision-Making

1. INTRODUCTION AND PRIOR WORK

1.1 Introduction

Public transportation agencies worldwide are increasingly leveraging data analytics and visualization tools to improve service reliability, transparency, and operational efficiency. IndyGo, the public transit agency of Indianapolis, aims to enhance its services through data-driven decision-making by developing an On-Time Performance (OTP) Dashboard using Power BI.

The OTP Dashboard addresses critical challenges in service reliability, route performance analysis, and decision support by visualizing transit operations data in an accessible format. The overarching goal is to establish a transparent and scalable

open data platform that provides real-time and historical insights into on-time performance trends, service adherence, and peak-hour analysis. This initiative aligns with the best practices of transit agencies nationwide, fostering accountability, efficiency, and innovation in urban mobility planning.

1.2 Prior Work

Several public transit agencies have developed performance dashboards to enhance service visibility and stakeholder engagement. Existing solutions include:

- MTA Bus Time (New York City): Provides real-time bus tracking and performance metrics to passengers and transit planners [1].

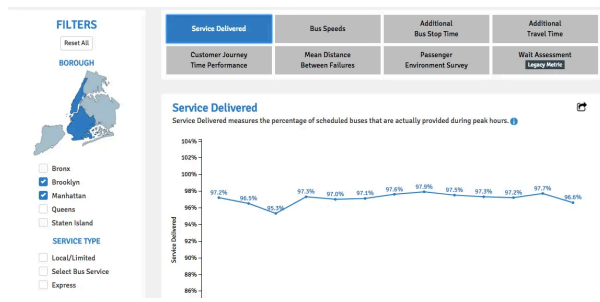


Fig. 2. MTA Dashboard

- Chicago Transit Authority (CTA) Performance Dashboard:** Displays monthly service metrics, including ridership trends and on-time performance [2].



Fig. 3. Chicago Transit Authority Performance Dashboard

- CapMetro Performance Dashboard:** Offers insights into route performance, safety statistics, and ridership analysis for Austin's transit network [3].

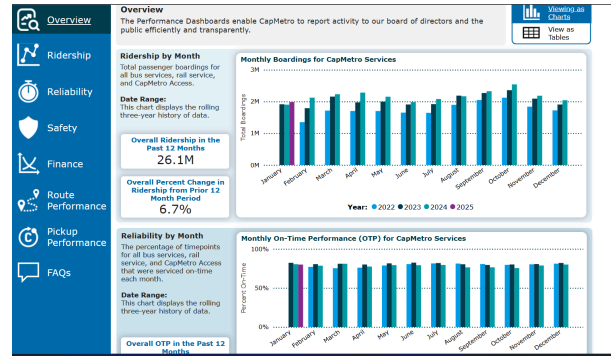


Fig. 4. CapMetro Dashboard

- Virginia Department of Rail and Public Transportation (DRPT) Performance Data Dashboard:** Provides quarterly updates on transit performance across Virginia, aiding in statewide transit planning and analysis [4].

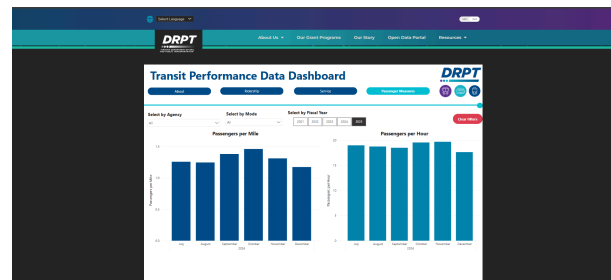


Fig. 5: Virginia Department of Rail and Public Transportation (DRPT) Performance Data Dashboard

- City of Phoenix Public Transit Performance Dashboard:** Part of Phoenix's City Manager's Performance Dashboard, which tracks ridership trends, service efficiency, and key transit metrics, aligning with the city's goals of transparency and continuous improvement [5].



Fig. 6. City of Phoenix Public Transit Performance Dashboard

- TransPro Consulting's TransDASH:** A national digital transit performance dashboard that shifts the industry from traditional ridership metrics to a more comprehensive approach, measuring the broader impact of public transit on communities [6].

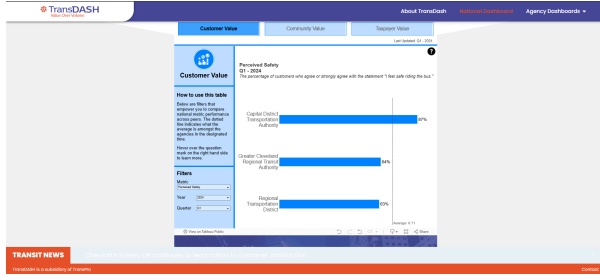


Fig. 7. TransPro Consulting's TransDASH Dashboard

While these platforms have successfully improved **transparency and operational analysis**, they exhibit certain limitations, such as **static reporting structures, limited interactive visualizations, and lack of predictive analytics**. The IndyGo OTP Dashboard expands upon these approaches by incorporating **real-time analytics, route-based performance breakdowns, and historical comparisons**, thus providing deeper insights for **strategic decision-making and service improvements**.

2. STAKEHOLDER GROUPS AND NEEDS

2.1 Stakeholder Groups

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2. STAKEHOLDER GROUPS AND NEEDS

2.1 Stakeholder Groups

The OTP Dashboard serves multiple stakeholders, each with distinct objectives:

- **Public Stakeholders:** Need real-time and historical access to transit data for transparency and accountability.
- **IndyGo Teams:** Require robust tools for analyzing service performance to guide operational improvements.
- **Transit Planners & Decision-Makers:** Need data-driven insights to optimize routes and evaluate policy changes.
- **Researchers & Data Analysts:** Utilize the platform for trend analysis, transit forecasting, and operational modeling.

- **Bus Operators:** Require insights into adherence trends and route-based scheduling optimizations.

2.2 STAKEHOLDER NEEDS

Each stakeholder group has unique data and visualization requirements:

- **Public & Transit Enthusiasts:** User-friendly, interactive dashboards to track bus performance trends.
- **Operations Team:** Live OTP metrics, route-specific delay patterns, and service reliability reports.
- **Planners & Decision-Makers:** Comparative route OTP trends, peak-hour congestion analysis, and policy impact assessments.
- **Researchers & Data Analysts:** Access to raw datasets, ridership trend models, and predictive analytics tools.

The dashboard is designed to balance these varying needs, ensuring that all stakeholders benefit from its insights while maintaining ease of use and intuitive data visualization.

3. DATA ACQUISITION

3.1 Data Sources

The dataset is provided by IndyGo and consists of GPS tracker and Automatic Passenger Counter (APC) data structured into fact and dimension tables covering the period from 2022 to 2024. The primary data sources include:

- **GPS Trackers:** Capturing bus location, arrival times, and deviations from schedules.
- **Automatic Passenger Counters (APCs):** Recording passenger boardings and alightings at each stop.
- **Schedule Data:** Comparing planned vs. actual service adherence.

3.2 Data Description, Quality, and Coverage

The dataset comprises dimension and fact tables, structured as follows:

Dimension Tables:

- **DimDate:** Contains date-related information, including the primary key referencing the trip date, date, and weekday type.
- **DimRoute:** Includes route details such as route key, route report label, route description, and active status.
- **DimStop:** Provides stop information with stop key, stop ID, and stop description.
- **DimBlock:** Contains block details with block key, block report label, and route description.
- **DimVehicle:** Includes vehicle details with vehicle key and vehicle report label.

- DimTrip: Offers trip details such as trip key, trip report label, direction, and trip farebox ID.
- DimUser: Contains user information with user key, logon ID, first name, and last name.

Fact Tables:

- FactSegmentAdherence (FSA): Includes adherence metrics for trip segments, such as departure and arrival times, passenger counts, and time variances.
- FactTimepointAdherence: Tracks on-time performance by recording scheduled vs. actual departure and arrival variances.
- FactHeadwayAdherence: Monitors bus spacing adherence to detect bunching or long service gaps.

4. DATA ANALYSIS

The OTP Dashboard will analyze data using:

- Descriptive Statistics: Identifying OTP trends over time.
- Comparative Analysis: Evaluating OTP variations across routes and service periods.
- Predictive Modeling: Forecasting potential delays based on historical patterns.
- Visualization Tools: Power BI dashboards for interactive data exploration.

Formulas Used -

```
% of Month OTP =
VAR TotalThisMonth =CALCULATE(COUNTROWS(indygo),ALL(indygo[OTP Category]),VALUES(indygo[MonthLabel]))
RETURN
DIVIDE(COUNTROWS(indygo),TotalThisMonth)
```

Fig. A DAX Formula for Calculating Monthly OTP Category Percentages

The percentage of Month OTP measure calculates the proportion of trips for each On-Time Performance (OTP) category relative to the total trips within the same month. It ignores category filters to compute an accurate monthly share for Early, On-Time, and Late trips. This allows for detailed monthly performance breakdowns across categories

```
PeakCategory =
SWITCH(
TRUE(),
HOUR(indygo[DepartTimeSchedule]) >= 6 && HOUR(indygo[DepartTimeSchedule]) < 10, "Peak",
HOUR(indygo[DepartTimeSchedule]) >= 15 && HOUR(indygo[DepartTimeSchedule]) < 21, "Peak",
"Non-Peak"
)
```

Fig. B. Logic to Classify Trips as Peak or Non-Peak Based on Departure Time

The PeakCategory measure classifies each trip as "Peak" or "Non-Peak" based on its scheduled departure time. Trips departing between 6–10 AM or 3–9 PM are labeled as "Peak", while all others are categorized as "Non-Peak." This enables

comparative analysis between peak and non-peak service performance

```
OTP Category =
SWITCH(
TRUE(),
indygo[DepartTimeVarianceSecs] < -60, "Early",
indygo[DepartTimeVarianceSecs] > 300, "Late",
indygo[DepartTimeVarianceSecs] >= -60 && indygo[DepartTimeVarianceSecs] <= 300, "On Time",
BLANK()
)
```

Fig. C. DAX Formula for Trip Classification Based on Departure Variance

The OTP Category measure classifies trips as "Early," "Late," or "On Time" based on their departure time variance in seconds. Trips departing more than 60 seconds early are marked "Early," more than 300 seconds late as "Late," and trips within -60 to +300 seconds as "On Time."

```
ServiceCategory =
SWITCH(
TRUE(),
indygo[TotalTimeScheduleSecs] <= 900, "Rapid", -- 15 minutes or less
indygo[TotalTimeScheduleSecs] <= 1800, "Frequent", -- 30 minutes or less
indygo[TotalTimeScheduleSecs] <= 3600, "Basic", -- 1 hour or less
indygo[TotalTimeScheduleSecs] > 3600, "Coverage", -- More than 1 hour
"Uncategorized"
)
```

Fig. D. DAX Formula for Trip Classification by Service Category Based on Scheduled Time

The ServiceCategory measure classifies trips based on scheduled travel time. Trips are categorized as "Rapid," "Frequent," "Basic," or "Coverage" depending on whether the scheduled time is under 15 minutes, 30 minutes, 1 hour, or more than 1 hour. This supports service-type performance analysis.

5. VISUALIZATIONS

The dashboard employs multiple visualization techniques to communicate insights effectively:

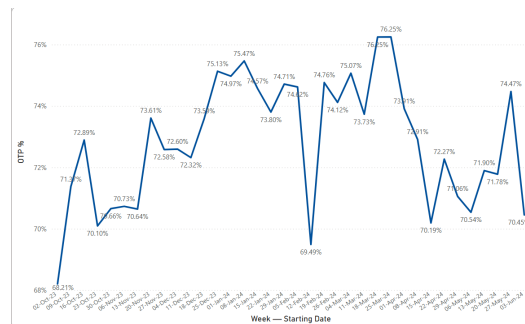


Fig. 8. Weekly Trend of Overall On-Time Performance (OTP %)

This line chart shows weekly changes in IndyGo's On-Time Performance (OTP %) from October 2023 to June 2024. Peak performance reached 76.25% in February, while a notable dip to 69.49% occurred later that month. The visualization

highlights service trends and supports data-driven operational improvements.

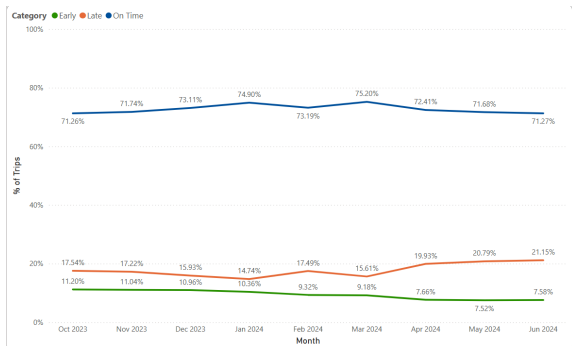


Fig. 9. Monthly Breakdown of On-Time, Late, and Early Trips (% of Trips)

This multi-line chart shows the monthly distribution of On-Time, Late, and Early trips from October 2023 to June 2024. On-time performance peaked at 75.20% in February, while late trips increased to 21.15% by June. The chart highlights shifts in service reliability and areas needing operational focus.

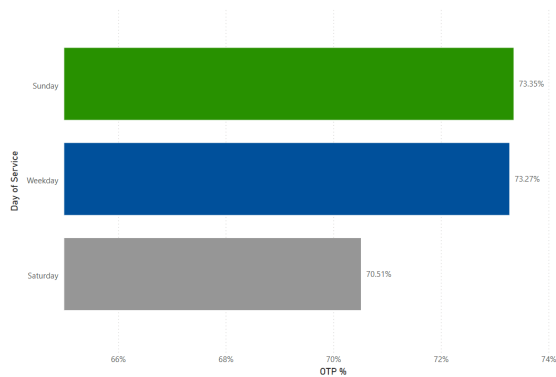


Fig. 10. On-Time Performance (OTP %) by Day of Service

This line chart shows IndyGo’s weekly OTP % from October 2023 to June 2024. Peak reliability was seen in February 2024 (76.25%), while dips, like in late February (69.49%), are clearly visible. It highlights periods of strong and weak service for operational improvements.

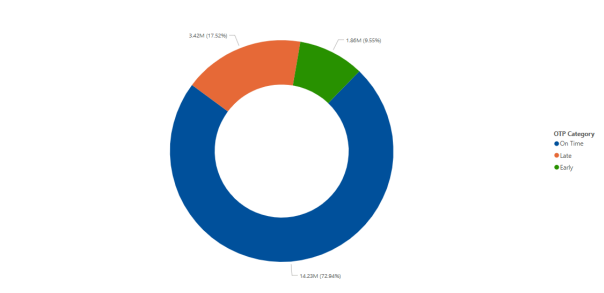


Fig. 11. Overall OTP Category Distribution – Donut Chart

This stacked bar chart compares peak and non-peak departure variances over four months. Peak time consistently shows higher delays than non-peak, with a noticeable gap each month, highlighting ongoing rush-hour impacts on service reliability.

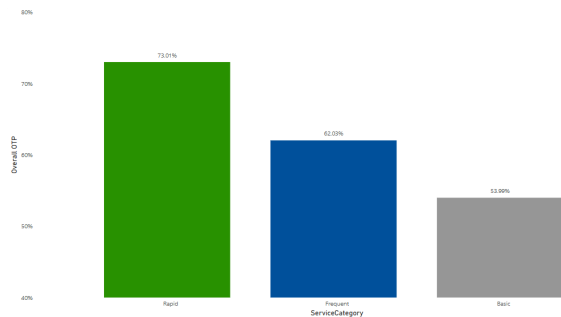


Fig. 12. On-Time Performance (OTP %) by Service Category

This bar chart compares OTP across service categories. Rapid services achieved the highest OTP at 73.01%, followed by Frequent services at 62.03%, and Basic services at 53.99%. It highlights stronger performance in higher-tier services.

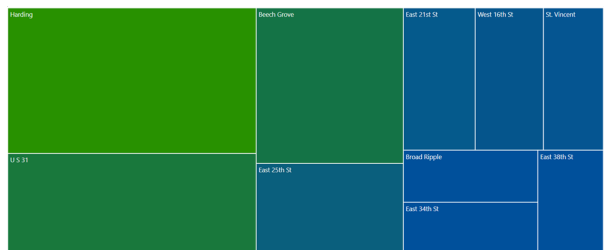


Fig. 13. Top 10 Routes by On-Time Performance (OTP %) – Treemap

This treemap shows the top 10 IndyGo routes based on their OTP performance. Larger green and blue blocks represent better-performing routes like East 16th St, Nora, and Broad Ripple. Gray blocks such as Mars Hill and East 21st St indicate comparatively lower OTP. The size and color help quickly identify top-performing and underperforming routes.



Fig. 12. Initial PowerBi Dashboard

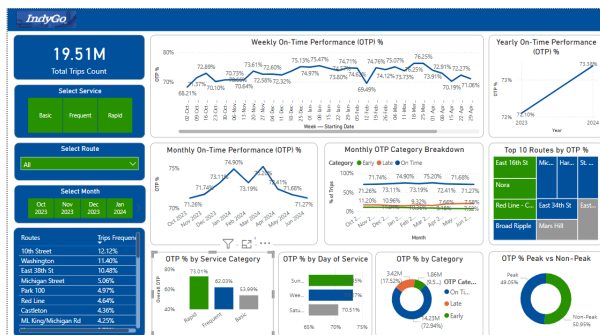


Fig. 14. Final IndyGo On-Time Performance (OTP) Dashboard Overview

The final IndyGo On-Time Performance (OTP) dashboard offers a clean and interactive layout that provides stakeholders with comprehensive insights into service reliability. It includes key performance indicators such as the total trip count (19.51M) and dynamic filters for Service Type, Route, and Month to enable customized exploration. The dashboard visualizes weekly, monthly, and yearly OTP trends, allowing users to monitor performance changes over time. OTP category breakdowns are presented through line and donut charts, highlighting the proportions of on-time, late, and early trips. Route-level performance is visualized using a treemap, while service category and day-of-service comparisons are shown through bar charts. Additionally, peak versus non-peak performance is analyzed through a separate chart. Overall, the dashboard improves usability with clear navigation, consistent visual styling, and optimized performance, ensuring that operational teams can easily identify trends, strengths, and areas for improvement in IndyGo's transit system.

6. Problems Surfaced During Evaluation and Redesign Solutions

During the evaluation, several challenges emerged that impacted the dashboard's initial performance and user experience:

Several challenges surfaced during the project that impacted dashboard development. The original dataset exceeded Power BI's 1GB upload limit; to address this, we dropped rows with null values and retained only essential columns, significantly reducing file size while preserving key insights. Data type preservation was another issue, as saving preprocessed files as CSV caused loss of datatype information, increasing processing time. Transitioning to the Parquet format maintained datatype integrity and optimized performance. Additionally, missing data spanning four months created gaps in the linear OTP trends. To mitigate this, we proposed creating separate Power BI files for individual visualizations and deploying them through iframes, alongside filling missing values using column means. Finally, the relatively short data timeframe (less than a year) was addressed by including both monthly and daily trend analyses, offering a more granular view of performance despite limited historical data.

7. Discussion of Challenges and Opportunities

Limitations and Issues Faced:

- Time Constraints:** Limited time necessitated prioritizing key preprocessing and visualization tasks while ensuring data accuracy.
- Data Quality:** The presence of missing data for several months required careful handling to maintain trend integrity and reliable analysis.
- File Size:** The large initial dataset forced us to significantly preprocess and filter the data to comply with Power BI's upload restrictions.

Complexity and Scaling Issues:

- Integrating multiple visualizations into a single Power BI file proved overwhelming for new users, as it complicated navigation and interpretation of graphs.
- Displaying comprehensive data within one report posed challenges in maintaining both performance and usability, emphasizing the need for a scalable solution.

Desirable Modifications and Extensions:

- **Modular Visualization Approach:**
Implementing separate Power BI files for distinct visualizations and deploying them via iframes will streamline navigation and ensure that each report remains under the 1GB limit.
- **Enhanced Navigation:**
Adding bookmark buttons to switch between specific OTP trend visualizations can further enhance user experience by making data exploration more intuitive.

Future Work and Promising Avenues:

Future enhancements could focus on incorporating granular data insights by analyzing individual OTP data for both arrivals and departures, offering more detailed performance metrics. Improved user guidance, such as adding descriptive text and tooltips alongside visualizations, would help new users interpret graphs more effectively and reduce confusion. Advanced missing data handling, including gap-filling using mean values, could optimize trend accuracy without distorting results. Additionally, computing and integrating missing labels like week numbers or bid periods could minimize misclassification risks. Refining tooltip and label formatting to display key metrics (e.g., "Avg Delay: 213 secs") would further enhance dashboard clarity. Finally, balancing multi-scale trends, such as visualizing both day-wise and month-wise patterns concurrently, remains a challenge that requires careful layout and labeling improvements to preserve readability without redundancy. This comprehensive redesign addresses the initial challenges by optimizing data processing, enhancing visualization interactivity, and ensuring a scalable, user-friendly experience, all critical for effective transit performance analysis.

8. RELATED WORK

Existing Transit Performance Dashboards:

- **MTA (NYC)** [1]: Real-time bus tracking.
- **Chicago CTA Dashboard** [2]: Route performance analysis.
- **CapMetro (Austin)** [3]: Ridership statistics and safety trends.

Critique of Existing Systems:

- **Strengths:** Enhanced public engagement, historical data availability.
- **Limitations:** Limited interactivity, lack of predictive analytics, less granular route-specific analysis.

The IndyGo OTP Dashboard builds upon these models by integrating real-time and historical performance tracking,

peak-hour trend identification, and geographic visualizations, ensuring a comprehensive transit performance analysis.

7. USAGE AND CRITIQUE OF AI TOOLS

The project explores AI tools such as:

- **Predictive modeling algorithms** for OTP forecasting.
- **Natural language query processing** to enable user-friendly dashboard interactions.
- **Limitations:** AI-based predictions may face **data bias, interpretability issues, and computational constraints**.

9. ACKNOWLEDGEMENTS

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